

Health data governance in the age of artificial intelligence: policy imperatives for the WHO European Region

Executive summary

Effective health data governance is essential for creating trusted, digitally enabled health systems that deliver high-quality, equitable care to populations. It underpins actions that support interoperability and sharing of data, enhance data quality and facilitate evidence-informed decision-making to lead to better health outcomes.

Health data governance also plays a crucial role in supporting artificial intelligence (AI) and machine learning in health, which is reliant upon access to large, high-quality datasets for training algorithms. In this context, health data governance ensures that data used for AI are ethically sourced, representative and free from bias, which helps in developing safe and reliable AI models that also enshrine gender, equity and human rights.

The 2022 WHO survey on digital health in the WHO European Region (referred to here as the 2022 WHO survey) highlighted several key trends and

challenges concerning the collection, use and sharing of health data. These findings are summarized in the report on digital health in the European Region and underscore the importance of building strong governance frameworks to harness the potential of health data while safeguarding individual privacy.

Main findings

Many Member States in the WHO European Region have enacted or updated national health data governance frameworks and strongly recognize the need for better data sharing and interoperability. Of the 98% of Member States in the Region that responded to the 2022 survey, 73% allocated special funding for bolstering digital health infrastructure during the coronavirus disease (COVID-19) pandemic, and 86% now legislate patient access to personal health data.

Findings from the 2022 WHO survey outlined in the report on digital health in the European Region are promising; however, significant challenges remain, particularly in the areas of data interoperability, standards adoption and stakeholder engagement. Health data are becoming increasingly complex, with sources ranging from electronic health records (EHRs) to social media and wearable devices, which pose both technical and ethical challenges. A key priority of countries is now to develop comprehensive legal frameworks that effectively balance the need for data sharing with the protection of individual rights.

The COVID-19 pandemic also highlighted the value of strong data governance, as countries with well-established frameworks were able to leverage large health datasets for real-time decision-making and pandemic response. Notably, the pandemic also exposed significant gaps in digital health systems, highlighting the need for sustained investment in digital health infrastructure, especially in regions where progress has been uneven.

Overview of main findings



86%

of Member States
legislate patient access
to personal health data

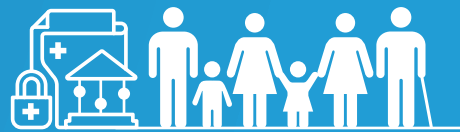


73%

allocated special funding for
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Health data governance
plays a crucial role in
supporting AI



Effective health data
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care to populations



Countries should invest in national
health data governance to improve
the quality of care, support
AI adoption and garner public trust





Urgent actions

Drawing from evidence gained through the 2022 WHO survey, this policy brief recommends four urgent actions for policy-makers to consider.

1

Further strengthen national health data governance to improve the quality of care, support AI adoption and garner public trust

Improve governance mechanisms to guide investment in secure digital health infrastructure, enhance data quality and establish legal and ethical guardrails for health data use

2

Develop a robust ecosystem of health data standards to facilitate the interoperability and exchange of data

Develop the national ecosystem for health data interoperability in support of standards adoption for primary and secondary health data exchange and use

3

Develop mechanisms or institutional arrangements to coordinate different national data providers

Establish mechanisms for the coordination of different data providers across the country to facilitate the collection, management and dissemination of consistent and complete data necessary for policy decision-making

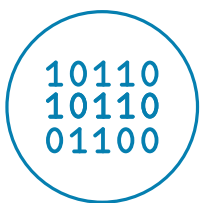
4

Involve patients, health-care providers and other key stakeholders in the development and implementation of health data governance frameworks

Engage stakeholders in policy design and build trust through inclusive governance processes that involve patients, health-care providers and other key stakeholders in the development of data governance frameworks supporting the implementation of trustworthy AI in health

By addressing the above actions, national health authorities can work towards establishing secure, digitally enabled health systems that help to facilitate care delivery, empower individuals and enable the potential of AI to be explored. Effective data

governance not only facilitates the responsible use of health data but also strengthens the overall capacity of the digital health landscape, enabling safe, high-quality health care for populations across the WHO European Region.



Introduction

Health systems globally have witnessed substantial growth in the scale and complexity of health data, encompassing a wide range of data types from clinical, biomedical, behavioural and social sources [1]. This ecosystem of health data presents both opportunities and challenges for health authorities, particularly in how to govern, secure and leverage these data to improve health outcomes.

High-quality health data are increasingly recognized as being vital to the delivery of integrated care across heterogeneous environments and to drive the development of global public health goods offering broad-scale benefit to populations [2]. Governance of these data is complex and is applied unevenly in Member States of the WHO European Region in terms of legal frameworks, technical infrastructure and stakeholder involvement. There is a need for coordinated efforts to establish data governance

frameworks that balance the rights of individuals with the potential collective benefits of health data use. The coronavirus disease (COVID-19) pandemic highlighted the importance of robust data governance, as countries with well-developed digital health systems were better equipped to utilize health data for rapid decision-making and pandemic response [3–5].

This brief is based on findings of the 2022 WHO survey as summarized in the report *Digital health in the European Region: the ongoing journey to commitment and transformation* [6]. It outlines the foundational aspects of health data governance that are necessary for establishing sustainable, trusted digital health ecosystems. These include legal frameworks, privacy protections, data-sharing protocols and mechanisms for building public trust.



Objectives

This policy brief aims to provide evidence-informed recommendations for developing and strengthening national health data governance frameworks to facilitate the safe and effective uptake of artificial intelligence (AI) into health systems.

As a minimum, these frameworks should:



set conditions that facilitate the collection, use and reuse of health data



ensure the security and integrity of health data stored within the health system through transparent governance, accountability and stakeholder engagement



balance the protection of individual privacy with the broader public health benefits of data use



support the ongoing development and innovation of digital health ecosystems by enabling data-driven health services, including AI, in alignment with WHO strategies



foster interoperability and data sharing across health systems





Key findings and analysis

Increased awareness of privacy and data protection

The 2022 WHO survey on digital health in the WHO European Region found that privacy protection is a key priority for countries in the Region (6). Many Member States have updated or implemented new legislation to strengthen privacy and data protection, and to ensure that individuals can exert greater agency and control over their personal health data (7–9), General Data Protection Regulation in the European Union (EU) serving as a model for many countries (10). The survey revealed that 86% of the Member States responding to the survey have legislated that patients should have access to their health data.

Response to lessons from the COVID-19 pandemic

The COVID-19 pandemic underscored the importance of having effective digital health systems and data governance frameworks in place. Countries with mature systems were able to rapidly mobilize and manage large health data sources and use them to inform public health action. The crisis also accelerated investments in digital health tools, with 73% of surveyed Member States allocating special funding for data-sharing and interoperability during the pandemic (4), leading to an unprecedented surge in the use of digital tools for remote health service provision.

Challenges in interoperability and data sharing

Despite the progress made, significant challenges remain, particularly in the areas of data interoperability and adoption of standards, with 98% of survey respondents reporting health information sharing and interoperability as national strategic priorities. Many countries still face difficulties in adopting and managing international standards for primary and secondary health data exchange and use. The lack of interoperability often results in suboptimal care and poorer treatment outcomes, and it can also significantly hinder the full use of health data for research, public health surveillance and health service delivery.

Data governance frameworks as conduits to developing trust for AI adoption in health

A data governance framework is the tangible expression of a country's social contract around data, allowing data governance to go from theory to practice (11, 12). Well-functioning data governance frameworks are essential for building trust in the use of health data. Such frameworks should include clear legal and ethical guidelines for data collection, use, storage and reuse. The Council Recommendation on Health Data Governance from the Organisation for Economic Co-operation and Development (OECD) provides valuable guidance for countries in establishing such frameworks (13).

The need for improved stakeholder engagement

Inclusion of diverse stakeholder perspectives, particularly those of patients, is critical for the success of health data governance frameworks. Engaging the public and other key stakeholders in the design and implementation of these frameworks is necessary to build trust and ensure that data governance mechanisms are transparent and responsive to societal concerns (14–16).



Policy considerations

Invest in health data governance to improve quality of care, support AI development and garner public trust

Investing in effective data governance and securing digital health infrastructures are essential to build public trust, enhance data quality and enable the responsible use of health data across various contexts, including care delivery, quality improvement, research and health statistics. The OECD's 12 high-level principles on health data governance can serve as a robust basis for building national health data frameworks (13,17).

The increasing use of AI across a variety of health contexts brings a new set of data governance considerations into focus as AI models are only as good as the data they are trained on. Development of successful AI systems relies on high-quality data for both training the algorithm and validating the algorithmic model (18). Consequently, poor-quality, biased or fragmented health data can lead to AI producing misdiagnoses, inaccurate predictions and harm to patients, in addition to exacerbating health disparities (19). Applying the FAIR (findability, accessibility, interoperability and reusability) data principles can help to mitigate bias in AI models by promoting transparency, reproducibility and inclusivity throughout the data lifecycle (20). Effective governance of data for AI also requires governments to aggregate and link data from across health and social care systems to create high-quality repositories for analysis by accredited researchers, with provision of secure analytics environments and/or with appropriate mechanisms for data extraction in place (21).

Data governance is also a core pillar of the European Health Data Space (EHDS), ensuring the secure exchange and ethical secondary use of health data. The EU has developed a comprehensive legislative framework that underpins responsible data governance and promotes high-quality, unbiased and privacy-compliant AI systems.

The EU Data Governance Act (22) fosters data sharing by establishing clear governance frameworks and mechanisms to enhance trust and transparency. Article 10 of the EU Artificial Intelligence Act (23) mandates strict data quality, bias mitigation and transparency requirements for high-risk AI applications. The Regulation on the EHDS (24) further strengthens secure, standardized and interoperable health data exchange across Member States while facilitating ethical data reuse. Aligned with the General Data Protection Regulation (25), these regulations collectively form a robust governance framework, ensuring AI in health care remains trustworthy, ethical and patient centred.

Strengthen the national ecosystem for health data standards to facilitate the interoperability and exchange of data

The 2022 WHO survey on digital health highlights the importance of interoperability as a strategic priority. Governments should work to strengthen the national ecosystem for health data standards to facilitate the seamless exchange of health data across different parts of the health system, and across geographical jurisdictions. Standards for data collection, exchange and use should be developed and adopted across all levels of the health system alongside certification and labelling processes that ensure quality and compliance.

Supporting mechanisms for coordinating diverse data providers across a country are also essential to facilitate the collection, management and dissemination of the consistent and complete data needed for effective policy decision-making.

Engage a variety of stakeholders in the development of health data governance frameworks

Effective governance frameworks require the inclusion of diverse stakeholder perspectives. Governments are uniquely positioned to lead this development, with the involvement of patients, health-care providers and other key stakeholders in framework design and implementation. Public consultation processes, such as surveys or participatory events, can help to ensure that these frameworks are responsive to the needs and concerns of the public.





Implementation considerations



Feasibility and resources

Implementing effective data governance frameworks requires significant investments in technical infrastructure, workforce capacity and regulatory oversight. Countries must allocate sufficient resources to support the long-term sustainability of digital health systems: not only to develop and maintain technical solutions but also to train health-care professionals and policy-makers in data governance principles and methods to leverage value from health data.



Stakeholders and partnerships

Collaboration among governments, international organizations, health-care providers, the private sector and patients is critical for the success of data governance initiatives. WHO can provide valuable support in the form of technical assistance, capacity-building activities and international stakeholder engagement.



Barriers and mitigation strategies

Regulatory and technological barriers, such as outdated legal frameworks or legacy data systems and repositories, must be addressed to enable the effective sharing and utilization of health data. Public concerns about privacy and data security should be mitigated through robust data protection and consent measures, and transparent communication [26,27].



Conclusions

The governance of health data is fundamental to advancing digital health and improving health outcomes across the WHO European Region. While the growing complexity and volume of health data present both opportunities and challenges for health systems, countries are urged to establish robust data governance frameworks that facilitate data sharing and balance privacy concerns with the public health benefits of data use. AI use brings a new set of data governance considerations into focus. Effective governance addresses how health data are collected, stored, shared and used by AI systems while maintaining trust and ethical standards. By investing in interoperability, engaging stakeholders and developing transparent data governance mechanisms, governments can build trusted, sustainable digital health ecosystems that provide equitable access to care, improve health outcomes and serve population health goals.





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